

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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Sec : 9

Week 3

Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>
int main()
{
    int pid;
    printf("Parent process started\n");
    printf("Parent PID: %d\n", getpid());
    pid = fork();
    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0)
    {
        printf("\nChild process created\n");
```

```
printf("Child PID: %d\n", getpid());  
printf("Parent PID: %d\n", getppid());
```

Output :

```
pothubhavana@DESKTOP-16SJJUA:~$ nano week3.c  
pothubhavana@DESKTOP-16SJJUA:~$ gcc week3.c  
pothubhavana@DESKTOP-16SJJUA:~$ ./a.out  
Parent process started  
Parent PID: 730  
  
Parent process is running  
Parent PID: 730  
Child PID: 731  
  
Parent state: WAITING  
Child process created  
Child PID: 731  
Parent PID: 730  
Child state: RUNNING  
Child state: TERMINATED  
Child process completed  
Parent state: RUNNING
```